

LUHANSK, Ukraine

WMO: 345230

Lat: 48.5656N

Lon: 39.2339E

Elev: 62

StdP: 100.58

Time Zone: 2.00 (EUE)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.0	(c) -17.5	(d) -23.7	(e) 0.4	(f) -20.5	(g) -20.6	(h) 0.6	(i) -17.1	(j) 7.5	(k) -4.2	(l) 7.0	(m) -5.7	(n) 1.4	(o) 280	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	35.0	20.6	33.0	20.2	30.9	19.5	22.2	30.5	21.3	29.7	20.6	28.6	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.5	25.1	18.7	13.7	24.3	17.9	13.0	23.6	65.2	30.1	62.2	29.7	59.5	28.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.2	5.4	-24.1	37.0	4.2	2.3	-27.1	38.6	-29.5	40.0	-31.8	41.2	-34.9	42.9
			-24.3	24.4	4.1	1.4	-27.2	25.4	-29.6	26.2	-32.0	27.0	-35.0	28.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-4.0	-3.4	2.1	10.2	16.4	20.6	23.1	21.6	15.6	9.1	2.9	-2.3
	DBStd	10.83	6.56	6.62	4.55	4.05	4.43	3.51	3.25	3.69	4.13	4.72	5.13	6.13
	HDD10.0	1787	436	376	246	46	6	0	0	0	5	73	218	382
	HDD18.3	3659	694	609	502	244	90	17	2	11	101	286	464	640
	CDD10.0	1566	0	0	2	52	204	317	406	360	174	47	4	0
	CDD18.3	398	0	0	0	1	31	83	150	113	19	2	0	0
	CDH23.3	4594	0	0	0	18	406	896	1664	1363	230	17	0	0
	CDH26.7	1899	0	0	0	2	121	342	755	618	58	2	0	0
Wind	WSAvg	2.3	2.4	2.7	2.9	2.6	2.0	2.0	1.8	1.9	2.0	2.0	2.4	2.6
Precipitation	PrecAvg	474	34	28	28	40	45	61	49	42	35	32	42	42
	PrecMax	693	92	87	72	140	89	164	145	104	122	95	90	102
	PrecMin	281	1	3	2	2	3	3	1	1	1	2	4	5
	PrecStd	95	24	21	16	26	24	37	38	31	28	24	22	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	10.7	17.2	25.8	32.6	34.7	37.6	38.5	30.9	25.9	17.3	11.0
		MCWB	7.2	7.7	10.2	14.8	19.0	20.8	21.0	21.1	17.5	17.0	12.5	9.3
	2%	DB	6.6	7.2	14.1	22.6	29.7	32.6	35.2	35.1	28.2	21.6	14.1	8.5
		MCWB	5.3	5.2	8.0	13.3	17.7	19.8	20.6	20.6	17.4	14.6	11.1	7.1
	5%	DB	5.0	5.2	11.2	20.1	27.7	30.5	33.1	32.6	26.0	18.8	11.6	6.4
		MCWB	4.0	3.9	6.9	12.0	17.0	19.2	20.3	20.2	16.7	13.9	9.5	5.3
	10%	DB	3.4	3.9	8.8	18.1	25.5	28.4	30.9	30.0	23.8	16.4	9.7	4.8
		MCWB	2.4	2.7	5.7	11.1	16.3	18.6	20.1	19.2	15.7	12.9	8.1	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	8.4	11.0	15.9	20.6	22.7	23.7	23.1	20.3	17.8	13.1	9.6
		MCDB	8.4	10.3	16.2	24.5	29.4	30.6	32.0	32.7	26.8	23.1	15.7	10.6
	2%	WB	5.2	5.4	8.9	14.2	19.1	21.3	22.4	21.8	18.8	15.7	11.2	7.1
		MCDB	6.2	6.6	12.5	20.6	26.8	28.9	30.2	31.2	24.8	19.8	13.7	8.4
	5%	WB	4.0	3.9	7.3	13.1	18.0	20.4	21.5	21.0	17.8	14.5	9.8	5.5
		MCDB	5.0	5.1	10.2	18.2	25.1	27.9	29.6	30.3	23.7	17.9	11.4	6.3
	10%	WB	2.5	2.8	5.9	12.0	17.0	19.5	20.8	20.0	16.7	13.2	8.2	3.9
		MCDB	3.3	3.8	8.6	16.7	23.9	26.5	28.9	28.0	21.7	15.9	9.6	4.6
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	7.7	11.1	12.5	12.0	12.4	13.6	12.3	9.2	6.3	5.1
		MCDBR	4.7	6.4	12.2	15.4	15.5	15.1	15.7	17.5	15.8	13.3	9.7	6.0
		MCWBR	4.1	5.0	7.3	7.7	6.2	5.5	5.1	6.2	6.9	7.2	6.8	5.0
	5% WB	MCDBR	4.6	6.0	10.5	12.5	13.9	13.0	13.2	14.5	12.8	11.1	8.6	5.9
		MCWBR	4.1	5.0	6.9	6.6	6.2	5.4	5.1	6.0	6.5	6.5	6.5	5.0
Clear-Sky Solar Irradiance	taub	0.333	0.365	0.397	0.459	0.422	0.419	0.445	0.446	0.415	0.381	0.354	0.337	
	taud	2.173	2.091	2.137	2.048	2.243	2.286	2.225	2.203	2.251	2.305	2.288	2.216	
	Ebn at Noon	693	755	802	788	841	845	815	794	779	741	672	635	
	Edh at Noon	83	113	128	155	132	128	134	131	114	91	75	71	
All-Sky Solar Radiation	RadAvg	0.99	1.94	2.99	4.33	5.80	6.27	6.12	5.59	3.88	2.21	1.02	0.63	
	RadStd	0.23	0.40	0.48	0.47	0.51	0.52	0.39	0.39	0.39	0.33	0.20	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.68	N/A	N/A	N/A	N/A	N/A	N/A	+164	+92	
(47) Regional (7 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-173	+117	+74

Nomenclature: See separate page